

Agro Trace Backend

MANUAL CHANGES GUIDE

For Running the Application

Blockchain-Based Agricultural Food Supply Chain Traceability

Smart India Hackathon (SIH) 2026

Version: 1.0 | Date: 16 August 2026

Build: JDK 21 + Spring Boot 3.4.0 + MySQL 8

0. IMPORTANT - Bugs Already Fixed

The following issues were found during verification and have ALREADY been corrected in the codebase. You do NOT need to change anything for these:

1. JWT Secret default was invalid base64 - would crash all logins. FIXED to a valid 38-byte key.
2. Database name mismatch between configs - aligned to agro_trace_dev.
3. Bouncy Castle dependency missing - added for Argon2id PIN hashing.
4. Lombok annotation processing not enabled - fixed with `<proc>full</proc>`.
5. FoodStandard version column collision with JPA `@Version` - renamed to `standard_version`.

1. REQUIRED CHANGES (Must do before running)

1.1 Create the MySQL Database

The application needs an empty database. Flyway will create all 35 tables automatically on first startup, but the database itself must exist.

MySQL CLI:

```
CREATE DATABASE IF NOT EXISTS agro_trace_dev;
```

OR use Docker (creates everything automatically):

```
docker-compose up -d
```

1.2 Set Database Credentials

The default username/password is `agro_trace_user` / `agro_trace_pass`. If your MySQL uses different credentials, override them when starting the app:

```
java -jar target/agro-trace-backend-1.0.0-SNAPSHOT.jar \  
  --DB_USERNAME=root --DB_PASSWORD=root \  
  --spring.profiles.active=dev
```

WARNING: If your MySQL root has no password, use `--DB_PASSWORD=` (empty) or create a dedicated user.

1.3 Run with the dev Profile

The dev profile enables the DemoDataSeeder which creates 7 sample users (PIN: 123456), 3 sample lots, test records, bundles, flags and complaints. Without it, you must register every user manually.

```
--spring.profiles.active=dev
```

2. OPTIONAL CHANGES (Recommended)

2.1 JWT Secret

The app now works out-of-the-box with the fixed default. Change it only for production. Generate a new one with:

```
# Linux / Mac:
openssl rand -base64 64

# Windows PowerShell:
[Convert]::ToBase64String((1..64 | ForEach-Object { Get-Random -Max 256 })))
```

Then set the environment variable or pass it at startup:

```
--JWT_SECRET=<your-generated-key>
```

2.2 Demo User PIN

All 7 demo users share the same 6-digit PIN (default: 123456). Change it in DemoDataSeeder.java line 71:

```
File: src/main/java/com/agro/trace/config/DemoDataSeeder.java
Line: user.setPinHash(passwordEncoder.encode("123456"));
```

2.3 Server Port

Default is 8080. Change in application.yml (or with --PORT=<port>):

```
server:
  port: ${PORT:8080}
```

2.4 Security Limits

Adjustable in application.yml under app.security:

```
app:
  security:
    pin-attempt-limit: 5           # PIN attempts before logout
    pin-lockout-duration-minutes: 30
    otp-length: 6
    otp-expiration-minutes: 10
    otp-rate-limit-per-mobile: 3
```

2.5 GPS Settings

```
app:
  qr:
    gps-required: true
    gps-max-accuracy-meters: 50
    geofence-enabled: false
```

2.6 Blockchain / Device / Simulation Flags

```
app:
  blockchain:
    enabled: false                # true = connect real Polygon Amoy node
  testing:
    simulated-mode: true         # false = use real device provider
  notification:
```

```
enabled: false
payment:
  enabled: false
```

3. Configuration File Reference

Where every important setting lives:

Setting	File	Line(s)
JWT secret	src/main/resources/application.yml	74
DB URL	src/main/resources/application.yml	6
DB username	src/main/resources/application.yml	7
DB password	src/main/resources/application.yml	8
Server port	src/main/resources/application.yml	43
PIN attempt limit	src/main/resources/application.yml	77
GPS settings	src/main/resources/application.yml	87-92
Blockchain flag	src/main/resources/application.yml	95-99
Demo user PIN	src/main/java/com/agro/trace/config/DemoDataSeed	71
Docker DB creds	docker-compose.yml	8-11, 30-33
Env template	.env.example	All

4. Demo Users (created by seeder in dev profile)

All demo users log in with PIN: 123456

Role	Login Identity	Role (RBAC)
Farmer	AADHAR-DEMO-FARMER	ROLE_FARMER
Agent	PF-COL-DEMO	ROLE_COLLECTING_AGENT
Supplier	PF-SUP-DEMO	ROLE_SUPPLIER
Manufacturer	PF-MFG-DEMO	ROLE_MANUFACTURER_EMPLOYEE
Distributor	PF-DIST-DEMO	ROLE_DISTRIBUTOR_EMPLOYEE
Retailer	AADHAR-DEMO-RET	ROLE_RETAILER
Government	PF-AG-DEMO	ROLE_GOVERNMENT_EMPLOYEE

5. Startup Commands Cheat Sheet

Option A - Local MySQL (recommended)

```
# 1. Create DB (once)
mysql -u root -p -e "CREATE DATABASE IF NOT EXISTS agro_trace_dev;"

# 2. Build
```

```
mvn clean package -DskipTests
```

```
# 3. Run
```

```
java -jar target/agro-trace-backend-1.0.0-SNAPSHOT.jar \
  --DB_USERNAME=root --DB_PASSWORD=root --spring.profiles.active=dev
```

Option B - Docker (all-in-one)

```
docker-compose up -d
```

```
# MySQL + app start automatically, DB auto-created
```

Access Points

```
API Base:      http://localhost:8080/api/v1
```

```
Swagger UI:    http://localhost:8080/api/v1/swagger-ui.html
```

```
Health Check:  http://localhost:8080/api/v1/actuator/health
```

6. Troubleshooting

Symptom	Solution
Login returns 500 / DecodingException	JWT secret invalid. Use the fixed default or set --JWT_SECRET with a base64 key (min 32)
Connection refused to MySQL	MySQL not running. Start it, or use docker-compose.
Unknown database agro_trace_dev	Create the DB first: CREATE DATABASE agro_trace_dev;
Access denied for user	Wrong DB credentials. Pass --DB_USERNAME and --DB_PASSWORD.
No demo users after login	Missing --spring.profiles.active=dev. DemoDataSeeder only runs in dev profile.
Lombok errors when building	Use Maven (not javac directly). <proc>full</proc> is configured in pom.xml.
Tests fail on Windows	Tests use H2 in-memory, no Docker needed. Run: mvn test
Flyway migration errors	Delete the database and let Flyway recreate it fresh (dev only).

Summary: Only 3 things are REQUIRED

1. MySQL running
2. Database agro_trace_dev created
3. Correct DB credentials

Everything else works with defaults. JWT secret bug already fixed.